Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Sequences & Series

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Geometric series: first term 5, common ratio $-\frac{2}{3}$. Find S_6 and S_∞. **[5]**

2. Geometric series: first term 8, common ratio $\frac{1}{2}$. Find S_8 and S_∞. **[5]**

3. Geometric series: first term 3, common ratio -0.4. Find S_5 and S_∞. **[5]**

4. Arithmetic series: $S_{20} = 710$, $u_{10} = 32$. Find a and d. **[5]**

5. Arithmetic series: $S_{15} = 285$, $u_5 = 17$. Find a and d. **[5]**

6. Arithmetic series: first term 4, common difference 3. Find S_{25}. **[4]**

7. Find the first three terms of $(1 - 2x)^5$. **[4]**

8. Find the first three terms of $(1 + 3x)^4$. **[4]**

9. Find the first three terms of $(2 - x)^6$. **[4]**

10. Find $\sum_{r=1}^{n}(3r+2)$. **[4]**

11. Find $\sum_{r=1}^{n}(2r-1)$. **[4]**

12. Find $\sum_{r=1}^{n}(4r+3)$. **[4]**

13. The binomial expansion of $(1+ax)^n$ begins $1+12x+66x^2+\cdots$. Find a and n. **[5]**

14. The binomial expansion of $(1+bx)^m$ begins $1+10x+40x^2+\cdots$. Find b and m. **[5]**

15. Show that $\sum_{k=0}^{n}(-\frac{1}{2})^k = \frac{2^{n+1}+(-1)^n}{3\cdot 2^n}$. Find S_∞. **[6]**

Section B – Hard (Q16–25)

16. Sequence: $u_1 = 3$, $u_{n+1} = 2u_n - 5$. Find closed form for u_n and $\lim u_n$. **[7]**

17. Sequence: $u_1 = 4$, $u_{n+1} = 3u_n + 2$. Find closed form for u_n. **[7]**

18. Sequence: $u_1 = 1$, $u_{n+1} = \frac{1}{2}u_n + 3$. Find closed form and limit. **[7]**

19. An arithmetic series has $S_n = 3n^2 + 5n$. Find the nth term. **[5]**

20. A geometric series has $S_n = 3(2^n - 1)$. Find the first term and common ratio. **[5]**

21. Find the sum of the first n terms of the series $5 + 9 + 13 + \cdots$. **[5]**

22. The sum of an infinite geometric series is 12 and the first term is 4. Find the common ratio. **[4]**

23. Find n if the sum of the first n terms of $2 + 5 + 8 + \cdots$ is 155. **[5]**

24. The third term of a geometric series is 12 and the sixth term is $\frac{3}{2}$. Find the first term and common ratio. **[6]**

25. The sum of the first 10 terms of an arithmetic series is 145 and the sum of the next 10 terms is 395. Find the first term and common difference. **[7]**

End of Worksheet • *Show all working clearly.*